

ST121 - 2026 - Statistical Laboratory

Assignment 2

Important information

The deadline for submission is **Thursday, 19th March (Week 10), 1pm.**

Please note that your report must be submitted in PDF. You should try and limit yourself to about 7 or 8 sides of A4, but without counting for any figures which may be needed. You will notice that a few questions (or part of questions) need to be answered analytically. You may want to answer them by hand. That is fine, you can scan your handwritten work and include it as image in your report for example. Just try to make sure the report is still well structured.

Also, please make sure you include your student ID code and lab group number and/or tutor name on the front sheet, but **NOT** your name!

In this initial marking scheme (I may amend it slightly), there will be a total of **60 marks** for this assignment, including **4** marks allocated to the writing layout and general presentation of your report. So make sure to be clear when answering, give the requested answers, explain by using sentences to link your reasoning, include appropriate justifications, label graphs, make the report nice to read for the marker... I will ask the markers to be fair but not too lenient either on this aspect.

Exercises

1. Anderson-Darling statistic

We saw in lectures how a QQ-plot or a probability plot can be useful to assess whether a sample is distributed according to a specific probability distribution. Although useful we would like to complete this graphical method with a statistical hypothesis test which would lead to a more objective and principled decision. Numerous tests have been proposed in the literature (in particular in order to test normality) and we focus here on the Anderson-Darling test. For an observed sample $x_1, \dots, x_n$ the Anderson-Darling (AD) test statistic is given by

$$T(x_1, \dots, x_n) = n \int_{-\infty}^{\infty} \frac{(F_n(y) - F_X(y; \theta))^2}{F_X(y; \theta)(1 - F_X(y; \theta))} f_X(y; \theta) dy$$

where $F_X(y)$ is the hypothesised cumulative distribution for the data, $f_X(y)$ is the corresponding probability density and

$$F_n(y) = \frac{\#\{i \in \{1, \dots, n\} : x_i \leq y\}}{n}$$

is the empirical distribution function of the observed sample.

- State, in words and at most two sentences, the null and alternate hypotheses in the present scenario.
- Briefly explain why the AD statistic may be useful to achieve our goal. In particular briefly comment on the roles played by the three terms, $(F_n(y; \theta) - F_X(y; \theta))^2$, $F_X(y; \theta)(1 - F_X(y; \theta))$ and $f_X(y; \theta)$.

- (c) While the expression above leads to an intuitive interpretation of what the statistic can achieve, a more useful expression is given by

$$T(x_1, \dots, x_n) = -n - \sum_{i=1}^n \frac{2i-1}{n} [\ln(F_X(x_{(i)}; \theta)) + \ln(1 - F_X(x_{(n+1-i)}; \theta))]$$

where $x_{(1)}, x_{(2)}, \dots, x_{(n)}$ is the order statistic of the sample. Most often the parameter(s) θ is/are unknown and must be estimated from the observed sample and t_{obs} can then be computed. From now on assume that we want to test whether a sample is drawn from a normal distribution, hence $\theta = (\mu, \sigma^2)^T$. The two datasets **x1** and **x2** referred to below can be found on Moodle, in the R file **samples_Q1**.

Write function `compute.ad.test(xs)` which takes in a vector of observations **xs** and returns the Anderson-Darling statistics. You should test your function on the two datasets **x1** and **x2**.

[Hint: the `ad.test` function in the `nortest` R library (which is not installed by default), may be a source of inspiration for your code and may be used to check that your own code produces plausible values (you will not get marks for using it but some of you may find it useful/reassuring).

- (d) To complete the statistical procedure we require computing the p -value. Even when θ is assumed known, the distribution of $T(X_1, X_2, \dots, X_n)$ under the null hypothesis is not tractable and it is unlikely that it will be when θ is estimated. The numerical method below works in both scenarios. This pseudo-code describes an algorithm based on simulation in order to compute the p -value for an observed statistic t_{obs} .

- Generate nB samples from F_X , and arrange them in $B = 10000$ groups of size n ;
- Calculate the AD test statistic for each sample of size n , leading to $T_1, T_2, \dots, T_n$.
- Compute the proportion of samples such that T exceeds T_{obs} , that is approximately the p -value with

$$p \approx \frac{\#\{b \in \{1, \dots, B\} : T_b \geq T_{\text{obs}}\}}{B}.$$

Write the R code corresponding to such pseudo-code to compute the p -values corresponding to **x1** and **x2**, assuming that the empirical mean and variance are used to estimate θ . For each of **x1** and **x2** plot the histogram of the simulated statistics and draw a vertical line for the position of the observed test statistic and on separate graphs plot the corresponding QQ-plots (you may use the functions `qqnorm` and `qqline`). Conclusions?

- (e) Explain in a few lines how you would adapt your code in order to test whether a sample is sampled from an exponential distribution. What is your conclusion about the generality of the approach?

[Total: 15 marks]

2. Standard Normal distribution

As said in the lectures, there remains one specific and quite important distribution which we need to know how to sample from, using only the $\sim U[0, 1]$ as sole source of randomness. This is the topic of this exercise.

It can be shown that $X_1, X_2 \stackrel{iid}{\sim} N(0, 1)$ iff their polar coordinates with $X_1 = R\cos(\theta)$, $X_2 = R\sin(\theta)$ are independent, with $\theta \sim U[0, 2\pi]$, $R^2 \sim \text{Exp}(\frac{1}{2})$.

Let U_1, U_2 two independent standard uniform distributions, i.e $U_1, U_2 \sim U[0, 1]$. We then define two independent random variables, X_1, X_2 , as

$$X_1 = R\cos(\theta), X_2 = R\sin(\theta),$$

with $\theta \sim U[0, 2\pi]$, $R^2 \sim \text{Exp}(\frac{1}{2})$.

- Using the Inverse-Transform method, write R in terms of U_1 , and θ in terms of U_2 .
- Hence use these to derive expressions for X_1, X_2 in terms of U_1 and U_2 .
- Write U_1, U_2 in terms of X_1 and X_2 .
- We now need to show that:

$$f_{X_1, X_2}(x_1, x_2) = \frac{1}{2\pi} \exp\left(-\frac{1}{2}(X_1^2 + X_2^2)\right)$$

by using an appropriate change of variables.

To prove that last equality, you will need the following result:

Suppose (W_1, W_2) has joint density $f(w_1, w_2)$ and let (Y_1, Y_2) be some function of (W_1, W_2) defined by $Y_1 = u_1(W_1, W_2)$ and $Y_2 = u_2(W_1, W_2)$ with the single-valued inverse given by $W_1 = v_1(Y_1, Y_2)$ and $W_2 = v_2(Y_1, Y_2)$.

The joint pdf Y_1 and Y_2 is given by

$$g(y_1, y_2) = f(v_1(y_1, y_2), v_2(y_1, y_2))|J|$$

where $|J|$ refers to the absolute value of the Jacobian, J . The Jacobian, J , is given by

$$J = \det \begin{pmatrix} \frac{\partial v_1(y_1, y_2)}{\partial y_1} & \frac{\partial v_1(y_1, y_2)}{\partial y_2} \\ \frac{\partial v_2(y_1, y_2)}{\partial y_1} & \frac{\partial v_2(y_1, y_2)}{\partial y_2} \end{pmatrix}$$

- From the expression of $f_{X_1, X_2}(x_1, x_2)$, explain why X_1 and X_2 are independent random variables.
- By using this method, use **R** to generate 10,000 independent realisations from $N(0, 1)$. Plot an histogram of your realisations, and add a line showing the pdf of a $N(0, 1)$ distribution.
- Call the `compute.ad.test(xs)` from part 1, and apply it to your vector of observations. You should confirm the nature of its distribution.
- Having generated samples from $N(0, 1)$ distribution, how would you now obtain a sample from a generic $N(\mu, \sigma^2)$ distribution?

You may need the univariate version of the previous result: Given W with pdf $f(x)$ and the transformation $Y = u(W)$ with the single-valued inverse $W = v(Y)$, then the pdf of Y is given by

$$g(y) = f(v(y))|v'(y)|.$$

- By using realisations from your $N(0, 1)$ distribution from (e), generate a sample of 10,000 realisations from a $N(2, 9)$ distribution. Draw its histogram.

[Total: 18 marks]

3. Exploratory Analysis and Hypothesis testing

In this section, you will be working on the `Tooth.Growth` dataset, available on Moodle. The response is the length of odontoblasts (cells responsible for tooth growth) in 40 guinea pigs. Each animal received one of two dose levels of vitamin C (1 and 2 mg/day) by one of two delivery methods, orange juice or ascorbic acid (a form of vitamin C and coded as `VC`). You then have a data frame with 40 observations and 3 variables:

- `len`: tooth length, numeric response;
- `supp`: supplement type, factor `VC` or `OJ`;
- `dose`: dose in milligrams per day, which we will treat as a factor.

(a) Overall distribution of `len`

- Represent the distribution of `len` with a graph, justifying your choice of graph type.
- Comment on the skewness and kurtosis of the distribution of `len`. Justify your answers.
- Comment on whether the distribution of `len` can be considered normal.

(b) `len` vs `supp`

- Draw two boxplots of `len`, one for each supplement type. Do the tooth length seem to be significantly affected by whether the vitamin C was delivered via orange juice or ascorbic acid?
- The answer to the previous question needs to be confirmed or infirmed statistically. Determine, with justification, which test needs to be used.
Note: do not perform any hypothesis test yourself here. You can “just” interpret some output following a relevant command call.

- Perform such hypothesis test at a 5% significance level to decide whether the tooth length varies depending on the supplement type. Make sure to include:

- the model assumptions
- the null hypothesis H_0 and the alternative hypothesis H_1 of the test, as well as justification for your choice of H_1 .
- the test statistic T
- the distribution of T under H_0
- the observed value t_{obs} of the test statistic
- the p -value of the test
- the critical region of the test
- the outcome and conclusion of the test.

Note: here you are expected to perform the test “manually” with R, not just interpret some output from a defined R function.

(c) `len` vs `dose`

- Draw two boxplots of `len`, one for each dose level. Do the tooth length seem to significantly increase when the amount of vitamin C increases?
- Same question as (b)(ii), this time with `dose` instead of `supp`.
- Same question as (b)(iii), this time with `dose` instead of `supp`.

[Total: 23 marks]

Total for the whole assignment: 60 marks